Cheat sheet

NLP

Classic NLP, tokenisation and embeddings

Pipeline

1 Text — 2 Normalise — 3 Tokenise —

4 Represent (BoW / TF-IDF / embeddings) — 5 Model — 6 Output

Concepts

Tokenisation	Word, subword (BPE, WordPiece, SentencePiece/Unigram), character
Stopwords	Very frequent words often removed in classic NLP
Stemming vs lemmatisation	Chop suffix vs dictionary base form
BoW / TF-IDF	Count vectors / weighted by rarity across documents
n-grams	Sequences of n tokens ("New York")
Word2Vec / GloVe / fastText	Static word embeddings (fastText handles sub-words)
Contextual embeddings	BERT-style: same word, different vector by context
Sentence embeddings	Sentence-Transformers, embedding APIs for search and RAG

Tasks

Classification	Sentiment, intent, topic, spam
NER	People, organisations, places, dates
QA	Extractive (span) vs generative
Summarisation	Extractive vs abstractive
Translation	Seq2seq / LLMs

Metrics

BLEU	n-gram overlap (translation)
ROUGE	Recall-oriented overlap (summaries)
BERTScore	Embedding similarity of outputs
Perplexity	Language-model quality

Pro tip

For modern generative tasks, n-gram metrics correlate poorly with quality; use rubric-based human or LLM-as-judge evaluation as well.